

# The Living Coherence

## Research Status Report on Its Logical and Formal Representation

Status as of 18 August 2026

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## Executive Summary

This report documents the current state of a project devoted to the logical and formal representation of **the living coherence** (der lebendige Zusammenhang). Its subject is a coherence in which actuality, structured potentiality, actual enactment, qualitative novelty, continuity, trace efficacy, and sustained transformability are related to one another. The report addresses this architecture alone. A theory of primordial consciousness is not part of the formalisation presented here.

The central advance consists in moving from individual forms to **typed configurations**:

$$V_{\text{act}}(e, \Gamma, d, \Gamma').$$

An actual enactment  $e$  takes a current configuration  $\Gamma$ , with the involvement of a direction  $d$  that is constrained but not fixed upon a completed target form, into an actual configuration  $\Gamma'$ . Transformation is not defined by difference alone, but by the conjunction of actual enactment, witnessed continuity, and qualitative novelty. A living enactment additionally requires ontological novelty and the sustained transformability of the configuration that has emerged.

The current state is a **formally consolidated draft**, but not yet a completed or mechanically verified theory. The types, level distinctions, and central definitions have been specified precisely enough to permit the construction of a complete axiom system and a non-vacuous minimal model. In particular, the following have not yet been presented:

1. a complete model of the corrected overall signature;
2. independence analyses and countermodels;
3. proofs of conservativity for the modules;
4. verification in Isabelle/HOL or by a comparable proof kernel;
5. a domain semantics that independently grounds  $Offen_{\text{ont}}$  and  $Neu_{\text{ont}}$ .

The theory can therefore already be presented as a **scientific research programme and formal specification**, but not yet as a proven universal ontology, a physical theory, or a completed theory of life.

**Translation convention.** The German expression lebendiger Zusammenhang is rendered as **living coherence**. Formal identifiers from the source signature—such as *Verankert*, *Realisierbar*, *Wandel*, *SchoepferischerWandel*, and *LebendigerVollzug*—are retained unchanged so that the English report remains cross-referable to the German formalisation. Their meanings are translated and explained in the surrounding text.

## Status Overview

| Area                             | Current status            | Evidential scope                                                                          |
|----------------------------------|---------------------------|-------------------------------------------------------------------------------------------|
| Definition of the subject matter | consolidated              | The object of inquiry and its core distinctions are clearly identified.                   |
| Typing                           | largely consolidated      | Configuration, direction, enactment, trace, quality, and context are distinct types.      |
| Core relations                   | consolidated draft        | Signatures and definitional dependencies are specified.                                   |
| Ontological openness             | explicit basic assumption | Formally represented, but not derived from logic or physics.                              |
| Continuity                       | viable schema             | Transformation with a criterion-bound witness; concrete criteria remain domain-dependent. |

| Area                    | Current status         | Evidential scope                                                 |
|-------------------------|------------------------|------------------------------------------------------------------|
| Event order             | formally corrected     | Acyclicity applies to event occurrences, not to form types.      |
| Modular architecture    | consolidated           | Core, extensions, specialisations, and metatheory are separated. |
| Minimal model           | pending                | Not yet fully documented for the corrected overall signature.    |
| Mechanised verification | pending                | No proof-kernel build of the present $T_{LZ}$ yet exists.        |
| Ontological adequacy    | open research question | It cannot be decided by model satisfiability alone.              |

# 1 Subject Matter and Aim

## 1.1 Guiding Question

The guiding question is:

How can a living coherence be represented logically in such a way that actuality, potentiality, and transformability are linked precisely, without reducing potentiality to a catalogue of completed future states, transformation to mere state change, or livingness to a physical metaphor?

The draft is intended to satisfy six requirements simultaneously:

1. **Actuality:** An actual enactment must be distinguished from a merely possible or describable continuation.
2. **Potentiality:** A direction must be anchored in an actual configuration without already containing a determinate result.
3. **Novelty:** Qualitative difference, theoretical non-derivability, and ontological non-predetermination must not be conflated.
4. **Continuity:** Transformation must contain a traceable continuation without requiring the identity of static properties.
5. **Trace and efficacy:** Earlier enactments must be able to continue operating in later configurations through typed traces, without every trace automatically being treated as a cause.
6. **Sustained transformability:** A living enactment must not necessarily close off the possibility of further formation.

## 1.2 Fundamental Methodological Decision

The project does not merge philosophies or physical theories into a single comprehensive doctrine. Instead, for each formal problem it asks which already developed concept or formalism supplies a suitable structure. The provenance is identified, but the adopted structure receives an independent definition within  $T_{LZ}$ .

Accordingly:

- Heisenberg, Dürr, Simondon, Peirce, Bohr, Einstein, Schrödinger, Leibniz, and others are **sources and points of comparison**, not themselves axioms.
- Quantum mechanics is a possible **domain instance**, not the ontological foundation of the core.
- Under specific conditions, the results of Gödel and Turing delimit formal theories; they do not prove ontological openness.
- Every extension is assigned an explicit status: definitional extension, conservative extension, specialisation, interpretation/metatheory, or domain instance.

## 2 Separation of Levels and Types

### 2.1 Three Levels

The architecture distinguishes at least three levels.

#### 2.1.1 Object Level

This level contains actual configurations, enactment events, directions, traces, and qualitative structures.

#### 2.1.2 Description Level

This level contains theories, configuration descriptions, admissible model extensions, and formal continuations. A description  $\rho$  is not an actual future form.

#### 2.1.3 Metatheory

This level contains statements about language, models, provability, satisfiability, conservativity, and computability. Tarski's distinction between object language and metalanguage, and the institution theory of Goguen and Burstall, are particularly relevant here.

The basic form is:

$$L_{LZ}, \quad T_{LZ}, \quad M \models \varphi.$$

Language, theory, model, and satisfaction relation are different objects. It follows in particular that:

$$\text{formal satisfiability} \neq \text{provability} \neq \text{ontological adequacy}.$$

### 2.2 Proposed Basic Types

| Type                | Meaning                                                                         |
|---------------------|---------------------------------------------------------------------------------|
| <i>Occ</i>          | concrete occurrence of a form or component                                      |
| <i>GType</i>        | form type; the same type may occur more than once                               |
| <i>Cfg</i>          | typed configuration                                                             |
| <i>Dir</i>          | formative direction without an encoded target configuration                     |
| <i>Event</i>        | actual enactment as an event occurrence                                         |
| <i>TraceCarrier</i> | carrier of a trace or continuing effect                                         |
| <i>TraceType</i>    | kind of trace, such as provenance, material continuation, or semantic reference |
| <i>Qual</i>         | qualitative or structural determination                                         |
| <i>Criterion</i>    | criterion of continuity                                                         |
| <i>Context</i>      | context of description                                                          |
| <i>Description</i>  | formal configuration description                                                |

*Cfg* should not be construed as a mere set of forms. A set loses multiplicities, roles, and internal relations. A typed multiset or an attributed hypergraph are possible suitable carrier structures.

## 2.3 Configuration Rather Than Individual Form

Earlier versions represented an enactment as

$$V(e, g, d, h).$$

This dyadic representation was too narrow. Transformation can be borne simultaneously by several components, traces, and conditions, and can alter more than one component. The corrected form is therefore:

$$V_{\text{act}}(e, \Gamma, d, \Gamma').$$

A newly emerged form within the target configuration can be marked separately:

$$\textit{Emergiert}(e, o, \Gamma').$$

In this way, the enactment remains an occurrence of the coherence as a whole without dissolving the distinguishable forms within the configuration.

## 3 Structured Potentiality

### 3.1 Potential Field

The potential field of a configuration is defined intensionally by constrained directions:

$$\Pi_{\mathcal{Z}}(\Gamma) = \{d \mid P_{\mathcal{Z}}(\Gamma, d) \wedge \textit{Verankert}_{\mathcal{Z}}(\Gamma, d) \wedge \textit{Realisierbar}_{\mathcal{Z}}(\Gamma, d)\}.$$

The three conditions perform different functions:

- $P_{\mathcal{Z}}(\Gamma, d)$ :  $d$  is a formative direction of the configuration.
- $\textit{Verankert}_{\mathcal{Z}}(\Gamma, d)$ :  $d$  arises from the configuration, its trace field, or a relational tension.
- $\textit{Realisierbar}_{\mathcal{Z}}(\Gamma, d)$ :  $d$  is not merely linguistically formulable but is supported by conditions of the coherence.

The precise semantics of  $\textit{Realisierbar}_{\mathcal{Z}}$  remains open. If it is defined solely through a consistent model extension, it describes only logical admissibility. An ontological reading requires a domain semantics of resources, conditions, or effective constraints.

### 3.2 Target-Openness of a Direction

The signature of  $P$  contains no target configuration:

$$P_{\mathcal{Z}} : Cfg \times Dir \rightarrow Bool.$$

This prevents the immediate identification of a direction with a target. It does not yet prove, however, that a model does not encode the result completely by indirect means. Logical and ontological openness are therefore separated.

### 3.3 Formal Transformability

Formal or model-theoretic transformability obtains when at least two non-isomorphic descriptive continuations are compatible with the same configuration and direction:

$$\begin{aligned} WF_{\log}(\Gamma) \iff \exists d, \rho_1, \rho_2 [d \in \Pi_{\mathcal{Z}}(\Gamma) \wedge \textit{Adm}_T(\Gamma, d, \rho_1) \\ \wedge \textit{Adm}_T(\Gamma, d, \rho_2) \wedge \rho_1 \not\cong \rho_2]. \end{aligned}$$

This definition witnesses underdetermination within a theory  $T$ . It does not assert that both descriptions exist as completed forms.

### 3.4 Ontological Transformability

$$WF_{\text{ont}}(\Gamma) \iff \exists d [d \in \Pi_{\mathcal{Z}}(\Gamma) \wedge \text{Offen}_{\text{ont}}(\Gamma, d)].$$

$\text{Offen}_{\text{ont}}$  states that the actual result is not completely predetermined by the actual initial conditions. At the present stage, this predicate is an **explicit ontological basic assumption**. It is not derived from a multiplicity of models, quantum indeterminacy, incompleteness, or a lack of knowledge.

This openness is the project's central outstanding burden of justification. A formalisation can investigate its consequences; it cannot establish its ontological validity by mere stipulation of symbols.

## 4 Actual Enactment and Describable Continuation

### 4.1 Actual Enactment

$$V_{\text{act}}(e, \Gamma, d, \Gamma')$$

states that  $e$  is an actual enactment in which  $\Gamma$  passes into  $\Gamma'$  with the involvement of  $d$ . The predicate is not a transition function. It therefore does not presuppose that every combination of  $\Gamma$  and  $d$  has exactly one result.

### 4.2 Description Level

$$\text{Adm}_T(\Gamma, d, \rho')$$

states only that, according to theory  $T$ , the result description  $\rho'$  is an admissible continuation. This separation prevents two errors:

1. An unrealised description is not treated as a simultaneously actual form.
2. The actual enactment is not reduced to the derivation of a theorem.

A later executable model may therefore employ rewrite steps, Petri-net transitions, or state transitions, but it must explicitly mark them as **representations** of enactments.

## 5 Continuity Through Transformation

### 5.1 Continuity Relation

Continuity is neither numerical identity nor equality of an intersection of properties. It is formulated as witnessed continuation:

$$K_c(e, \Gamma, \Gamma') \iff \exists q, q', \varphi [\text{Traegt}(\Gamma, q) \wedge \text{Traegt}(\Gamma', q') \wedge \text{Fortfuehrung}_c(e, q, q', \varphi)].$$

$\varphi$  is the transformation witness. The criterion  $c$  specifies which kind of continuation is relevant. Possible cases include:

- preservation of a function through material change;
- transformation of a structure under an admissible map;
- continuation of a problem or organisational role;
- preservation of an invariant;
- controlled change of an equivariant.

## 5.2 Non-Circularity

A criterion of continuity must not be chosen arbitrarily only after the result is known. A schema of the following form is required:

$$Zulaessig(c, e, \Gamma) \implies Verankert(c, \Gamma, \Sigma_{<e}) \wedge Unterscheidungsfaehig(c).$$

The criterion must therefore be anchored in the initial configuration, its history, or the enactment type, and it must be capable of distinguishing continuation from rupture.

## 5.3 Einstein and Noether

Einstein's theories of relativity do not supply an ontology for  $T_{LZ}$ , but they do offer a methodological motif: descriptions should be covariant under admissible changes of representational frame. One possible schema is:

$$Gilt(q, g, \kappa) \iff Gilt(Tq, Tg, T\kappa)$$

for an admissible context isomorphism  $T$ .

Noether's work provides a powerful model for the relation between symmetry and invariance. Her physical-mathematical theorem, however, applies to variational problems with the relevant symmetries. The Noether theorem is therefore not universalised for the general core; only the broader research principle of examining  $K_c$  in terms of invariants and equivariants is adopted.

# 6 Novelty

## 6.1 Three Concepts of Novelty

The theory distinguishes the following.

### 6.1.1 Qualitative Novelty

$$Neu_{\text{qual}}(e, \Gamma, \Gamma')$$

obtains when  $\Gamma$  and  $\Gamma'$  are non-isomorphic relative to the relevant qualitative signature:

$$\Gamma \not\cong_Q \Gamma'.$$

### 6.1.2 Theory-Relative Novelty

$$Neu_T(e, \Gamma, d, \rho')$$

obtains when the description  $\rho'$  cannot be derived from the initial information formalised in  $T$ . Here  $T$  is a metatheoretical index.  $Neu_T$  must not be identified with ontological novelty.

### 6.1.3 Ontological Novelty

$$Neu_{\text{ont}}(e, \Gamma, d, \Gamma')$$

states that the concrete determination of  $\Gamma'$  was not completely predetermined by the actual initial conditions.

The following is a desired relation:

$$Neu_{\text{ont}} \implies Neu_{\text{qual}}.$$

The converse does not hold. A qualitatively different result may be completely determined.



## 6.2 Formal Boundary

Neither Gödel's incompleteness nor Turing's undecidability proves  $Neu_{ont}$ . Gödel's result concerns sufficiently strong, consistent, and effectively axiomatised formal theories. Turing undecidability presupposes an effective encoding and a concrete reduction. Both concern properties of formalisms and computational problems, not immediately the ontological predetermination of an actual enactment.

## 7 Traces, Efficacy, and Event Order

### 7.1 Typed Trace Field

The trace field of an emerged configuration is schematically written as

$$\Sigma_e(\Gamma') = \{(o, r, \tau) \mid Spur_\tau(r, o, e, \Gamma')\}.$$

Here:

- $o$  is a concrete occurrence or source;
- $r$  is a trace carrier;
- $\tau$  is a trace type;
- $e$  is the mediating enactment.

At least the following kinds of trace should be distinguished:

1. **Provenance trace:** Where did a contribution originate?
2. **Material or operational continuation:** What continues to operate in the new enactment?
3. **Evidence or record trace:** What bears witness to an earlier enactment?
4. **Semantic trace:** To what does a present structure refer?

Causal efficacy does not follow automatically from a provenance trace. A later causal semantics must supply additional rules or intervention criteria.

### 7.2 Recontextualisation

$$Rekontextualisiert(e', r, \tau, \tau')$$

states that the same trace carrier  $r$  acquires a different role in a later enactment. Its origin is not thereby altered retroactively. This distinction permits a change of meaning without backward causation.

### 7.3 Acyclicity Only at the Level of Event Occurrences

An earlier version imposed acyclicity upon the trace relation between forms. This was too strong. Living organisations can repeat forms, constitute feedback loops, or operate cyclically.

It is therefore the precedence relation between concrete events that is strictly partially ordered:

$$\prec_E \subseteq Event \times Event.$$

Irreflexivity and transitivity are required:

$$\neg(e \prec_E e), \quad e_1 \prec_E e_2 \wedge e_2 \prec_E e_3 \implies e_1 \prec_E e_3.$$

Concurrency can be defined in an event structure by

$$e_1 \parallel e_2 \iff \neg(e_1 \preceq_E e_2) \wedge \neg(e_2 \preceq_E e_1) \wedge \neg(e_1 \# e_2),$$

where  $\#$  denotes conflict. This structure is well prepared by Winskel and, as a related ordering principle, by Lamport's happened-before relation.

## 8 Composite Core Concepts

### 8.1 Transformation

$$\begin{aligned} Wandel_c(e, \Gamma, d, \Gamma') &\iff V_{\text{act}}(e, \Gamma, d, \Gamma') \\ &\quad \wedge K_c(e, \Gamma, \Gamma') \\ &\quad \wedge Neu_{\text{qual}}(e, \Gamma, \Gamma'). \end{aligned}$$

This definition requires an actual enactment, traceable continuity, and qualitative change. It does not yet presuppose ontological openness.

### 8.2 Creative Transformation

$$\begin{aligned} SchoepferischerWandel(e, \Gamma, d, \Gamma') &\iff Wandel_c(e, \Gamma, d, \Gamma') \\ &\quad \wedge Neu_{\text{ont}}(e, \Gamma, d, \Gamma'). \end{aligned}$$

### 8.3 Living Enactment

$$\begin{aligned} LebendigerVollzug(e, \Gamma, d, \Gamma') &\iff SchoepferischerWandel(e, \Gamma, d, \Gamma') \\ &\quad \wedge WF_{\text{ont}}(\Gamma'). \end{aligned}$$

Here, livingness denotes not merely change, but a transformation that brings forth actual novelty and preserves the transformability of the configuration that has emerged.

### 8.4 Two Strengths of Livingness

Minimal livingness:

$$Lebendig_{\text{min}}(\mathcal{Z}) \iff \exists e, \Gamma, d, \Gamma' \text{ LebendigerVollzug}(e, \Gamma, d, \Gamma').$$

Sustained livingness:

$$\begin{aligned} Lebendig_{\text{fort}}(\mathcal{Z}) &\iff \exists \Gamma_0 \text{ Aktiv}_{\mathcal{Z}}(\Gamma_0) \\ &\quad \wedge \forall \Gamma [AktErreicht_{V_{\text{act}}}^*(\Gamma_0, \Gamma) \rightarrow WF_{\text{ont}}(\Gamma)] \\ &\quad \wedge \exists e, \Gamma, d, \Gamma' \text{ LebendigerVollzug}(e, \Gamma, d, \Gamma'). \end{aligned}$$

$AktErreicht^*$  refers only to events that have actually been enacted. The definition does not presuppose an already existing infinite future.

## 9 Modular Architecture

### 9.1 Core and Semantic Modules

| Module                  | Function                                                 | Principal sources                                         | Status                                |
|-------------------------|----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------|
| $T_{LZ}^0$<br>$T_{POT}$ | typed core<br>structured,<br>constrained<br>potentiality | independent draft<br>Heisenberg as an<br>interpretive aid | consolidated draft<br>semantic module |

| Module    | Function                              | Principal sources | Status                                                           |
|-----------|---------------------------------------|-------------------|------------------------------------------------------------------|
| $T_{EFF}$ | efficacy and relational continuation  | Dürr              | semantic module; relations still require further differentiation |
| $T_{GEN}$ | genesis of directions from tensions   | Simondon          | highly relevant                                                  |
| $T_{MED}$ | mediation, habit, recontextualisation | Peirce            | highly relevant                                                  |

## 9.2 Formal Modules

| Module      | Function                                                       | Suitable formalisms                              | Status                         |
|-------------|----------------------------------------------------------------|--------------------------------------------------|--------------------------------|
| $T_{COMP}$  | event order and process composition                            | event structures, Lamport orderings, hypergraphs | priority                       |
| $T_{INV}$   | invariance and continuity witnesses                            | context isomorphisms, invariants, equivariants   | priority                       |
| $T_{CTX}$   | context-indexed validity                                       | indexed logic; later presheaves/sheaves          | basic module viable            |
| $T_{EXEC}$  | executable representation                                      | rewriting logic, coloured Petri nets             | implementation candidate       |
| $T_{MODAL}$ | model-theoretic openness                                       | Kripke or neighbourhood semantics                | $WF_{\log}$ only               |
| $T_{CAUS}$  | causality and intervention                                     | structural causal models                         | later specialisation           |
| $T_{RES}$   | resource modes                                                 | linear/affine logic, subexponentials             | later specialisation           |
| $T_{META}$  | satisfaction, translation, and limits of proof and computation | Tarski, institutions, Gödel, Turing              | indispensable protective layer |

## 9.3 Composition: Correction of the Operadic Proposal

Ordinary operads typically have multiple inputs and one output. An enactment  $\Gamma \rightarrow \Gamma'$ , however, may have several components on both sides. There are therefore two possibilities:

1. The entire target configuration  $\Gamma'$  is treated as a single output type. A coloured operad may then suffice.
2. Components are to remain individually composable and open interfaces are to remain visible. PROPs, properads, symmetric monoidal categories, multicategories, or structured cospans are then more suitable.

Hypergraphs represent polyadicity and provenance, but they establish neither causality nor physical non-locality. A corresponding representation theorem must show which core relations are preserved by the chosen structure.

## 9.4 Executable Representation

Rewriting logic fits transitions of the form

$$[\Gamma] \xrightarrow{e,d} [\Gamma'].$$

Coloured Petri nets are suitable for typed resources, concurrency, synchronisation, and reachability analysis. Neither formalism can by itself decide whether a modelled step is an ontologically open enactment. They provide an operational semantics for  $WF_{\log}$  and  $V$ -structures.

## 10 Assessment of the Philosophical and Physical Sources

### 10.1 Heisenberg

Heisenberg's recourse to Aristotle's *potentia* is relevant to a semantics of structured potentiality. What is adopted is the distinction between actual reality and a possibility that is not merely subjective. What is not adopted is the claim that the living coherence is grounded in the formalism of quantum mechanics.

**Status:** high conceptual relevance; no formal or physical foundation of  $T_{LZ}$ .

### 10.2 Hans-Peter Dürr

Dürr's emphasis on efficacy, relation, and wholeness supports  $T_{EFF}$ . His language, however, is not a completed formal theory. Concepts such as efficacy, relation, and potentiality must be independently typed and supplied with inference rules in  $T_{LZ}$ .

**Status:** high ontological inspirational value; formal reconstruction required.

### 10.3 Simondon

Simondon's individuation, metastability, and transduction address a central gap: how can a direction arise from tensions within a coherence without already containing a target form?

One possible schema is:

$$Generiert_{\mathcal{Z}}(\Gamma, \Sigma, s, d).$$

**Status:** one of the strongest additions to  $T_{GEN}$ .

### 10.4 Peirce

Peirce's triadic mediation, thought of continuity, and account of habit formation are relevant to typed traces, recontextualisation, and the stabilisation of recurrent enactment patterns.

**Status:** one of the strongest additions to  $T_{MED}$ ; adoption of his entire metaphysics is not required.

### 10.5 Bohr

Bohr's complementarity concerns context-dependent conditions of description. It must not be transferred to arbitrary pairs of concepts. In particular,  $K_c$  and  $Neu$  are not complementary, because a transformation requires both at once. Nor are trace order and trace field incompatible in principle.

For  $T_{CTX}$ , by contrast, the following is useful:

$$Gilt(q, g, \kappa).$$

Complementarity can be defined between contexts of description for which no complete common refinement exists, but which jointly cover a question.

**Status:** relevant as a contextual semantics, not as an ontology of the core.

## 10.6 Einstein

Einstein's special theory of relativity motivates the absence of a privileged global simultaneity in a physical instance. The general idea of covariance supports  $T_{INV}$ . It does not follow, however, that every perspective is equivalent or that every living coherence possesses the structure of relativistic spacetime.

General covariance alone is also not identical with complete background independence. The trace field must be identified neither with a metric nor, without further proof, with a topology.

**Status:** high relevance for representational invariance; physical causal structure only as a domain instance.

## 10.7 Schrödinger

In a quantum instance, Schrödinger's analysis of entangled systems establishes the possibility of non-factorisable total states:

$$\rho_{\Gamma} \neq \bigotimes_i \rho_i.$$

This is not a universal statement about all configurations of  $T_{LZ}$ . Nor does the Schrödinger equation ground ontological openness, because unitary dynamics is deterministic.

The "aperiodic crystal" in *What Is Life?* is relevant as a historical example of a stable biological trace carrier. Thermodynamic concepts, however, belong in  $T_{BIO}$  or  $T_{DISS}$ .

**Status:** high relevance in  $T_Q$ , supplementary relevance in biological specialisations, but not part of the core.

## 10.8 Leibniz

Leibniz's relational account of space and time fits the internal order of the coherence. The concept of compossibility is additionally fruitful. It can describe the joint compatibility of several directions:

$$Kompossibel_{\Gamma}(D_0) \iff \exists \mu \forall d \in D_0 \text{ Kompatibel}_{\Gamma}(\Gamma, d, \mu).$$

$\mu$  designates a joint descriptive continuation, not an already existing possible world.

Not adopted are the complete concept of an individual, pre-established harmony, windowless monads, and a universal identity of indiscernibles. These assumptions would endanger actual mediation or ontological novelty.

**Status:** high relational and modal relevance; deterministic components function as a contrasting case.

## 10.9 Noether

Noether's theorems connect continuous symmetries of variational problems with conservation statements. They are not universalised for  $T_{LZ}$ . What is relevant is the more precise question of which structures or relations pass through a transformation invariantly or equivariantly.

**Status:** very high relevance to the elaboration of  $K_c$ ; a full transfer of Noether's theorem only in suitable special models.

## 10.10 Gödel and Turing

Gödel delimits sufficiently strong effective formal theories. Turing delimits algorithmic decision procedures. For  $T_{LZ}$ , both serve as metatheoretical safeguards against unfounded claims of completeness.

The following inferences are inadmissible:

$$\text{Goedel} \Rightarrow \text{Offen}_{\text{ont}},$$

$$\text{Turing} \Rightarrow \neg \text{Entscheidbar}(WF),$$

unless concrete assumptions, encodings, and reductions are supplied.

**Status:** indispensable in  $T_{META}$ , but without ontological evidential force.

## 11 Specialisations and Domain Instances

### 11.1 Quantum Instance $T_Q$

$T_Q$  may contain state spaces, observables, contextuality, entanglement, and fermionic exclusion conditions. Pauli supplies no universal axiom of individuality. Entanglement supplies no universal non-separability of the living coherence.

### 11.2 Dissipative and Biological Instances

Prigogine's dissipative structures presuppose defined thermodynamic quantities, flows, boundaries, and non-equilibrium conditions. The following therefore does not hold universally:

$$\text{Stabilisierung} \Rightarrow \text{permanenter Neuheitsfluss}.$$

Maturana and Varela provide a specialisation of organisational closure: a network recursively produces its components and the conditions of its unity. Autopoiesis is thus a possible class of living organisation, but it is not definitionally identical with the general living coherence.

### 11.3 Social and Communicative Instances

In  $T_{SOC}$ , Luhmann's operational closure and structural coupling may be used to investigate how effects are reconstructed in the target system. The semantics of social theory must not be elevated to the universal core.

### 11.4 Coalgebra and the $\pi$ -Calculus

Coalgebras are suitable for observable behaviour and bisimulation. A coalgebra that contains successors, however, represents a formal space of possibilities and not necessarily ontological potentiality that remains open with respect to its target.

The  $\pi$ -calculus can model mobile communication structures. It is not an independent formalism of autopoiesis and is relevant only if dynamically changing connection structures become part of a domain instance.

## 12 Status of Extensions and Proofs

### 12.1 Five Kinds of Extension

1. **Definitional extension:** New symbols are completely defined in terms of old ones.
2. **Conservative extension:** No new sentences in the old language are generated.
3. **Specialisation:** Additional axioms deliberately restrict the class of models.
4. **Interpretation or metatheory:** A theory is translated into another theory or examined from outside.

5. **Domain instance:** The abstract core is interpreted by physical, biological, or social structures.

A model-conservative extension requires:

$$\forall M \models T_{LZ}^0 \exists M' \models T_{LZ}^0 \cup T_X : M' \upharpoonright_{\Sigma_{LZ}} = M.$$

This has not yet been proved in general for any substantive additional module. In particular, event structures, operads, or sheaves are not conservative merely because they “only represent” existing concepts. In every case, an explicit interpretation or a representation theorem is required.

## 12.2 Current Level of Maturity

For clear communication, the following scale is proposed:

| Level | Meaning                                       | Status of $T_{LZ}$                       |
|-------|-----------------------------------------------|------------------------------------------|
| S0    | philosophical intuition                       | surpassed                                |
| S1    | typed signature and separation of levels      | largely achieved                         |
| S2    | consolidated definitions and dependencies     | achieved for the core                    |
| S3    | non-vacuous model and countermodels           | pending for the corrected overall theory |
| S4    | mechanised proofs and builds                  | pending                                  |
| S5    | domain interpretation and adequacy assessment | pending                                  |

The accurate overall status is therefore: **S2, in transition to S3.**

## 13 What Can Currently Be Claimed with Confidence

### 13.1 Formally Robust Statements

1. The distinction between actual enactment and possible description is logically well-defined and necessary.
2. Configurations are a more appropriate primary bearer of potentiality and transformation than isolated individual forms.
3.  $WF_{\log}$  and  $WF_{\text{ont}}$  are not equivalent.
4.  $Neu_{\text{qual}}$ ,  $Neu_T$ , and  $Neu_{\text{ont}}$  are distinct concepts.
5. Continuity can consistently be modelled as transformation with a criterion-bound witness.
6. Event precedence can be represented without an external linear world time by means of a strict partial order.
7. Acyclicity must apply to event occurrences; form types and organisational forms may recur.
8. Efficacy, provenance, evidence, and semantic traces must be typed.
9. Physical theories, process formalisms, and metatheorems require clearly identified interpretation bridges.

### 13.2 Statements Not Yet Proven

1. That reality is ontologically open.
2. That every actual coherence is transformable.
3. That the complete corrected axiomatisation is consistent.
4. That all intended modules are conservatively extendable.

5. That  $WF_{\text{ont}}$  or  $Neu_{\text{ont}}$  is formally decidable or undecidable.
6. That the core adequately describes a biological, physical, or social domain.
7. That the proposed definition captures all and only living coherences.

## 14 Explicitly Excluded Fallacies

In particular, the following do **not** follow from the present system:

- quantum formalism  $\Rightarrow$  universal ontology;
- Heisenberg's uncertainty relation  $\Rightarrow$  free or creative formation;
- Schrödinger entanglement  $\Rightarrow$  universal holism;
- Pauli exclusion  $\Rightarrow$  a general principle of individuality;
- correlation without a trace path  $\Rightarrow$  acausality or synchronicity;
- hyperedge or coherence  $\Rightarrow$  Bell non-locality;
- Einstein covariance  $\Rightarrow$  equivalence of all perspectives;
- Gödel or Turing  $\Rightarrow$  ontological openness;
- linear logic  $\Rightarrow$  physical consumption of every trace;
- sheaf theory  $\Rightarrow$  an invariably existing global unity;
- minimal model  $\Rightarrow$  ontological truth;
- autopoiesis  $\Leftrightarrow$  every living coherence.

## 15 Comparative Assessment

### 15.1 Strengths in Comparison with Purely Philosophical Process Ontologies

The draft is stronger than a merely metaphorical process description in the following respects:

- explicit typing;
- configuration-level analysis rather than linear individual states;
- separation of actual and describable continuations;
- separation of three concepts of novelty;
- non-circular witnessing of continuity;
- explicit status logic for extensions;
- clear limitation of physical analogies;
- compatibility with formal verification.

### 15.2 Current Weaknesses in Comparison with Established Formal Theories

In comparison with mature formal theories, the following are missing:

- a final signature file;
- fully quantified axioms;
- a non-vacuous model of the current version;
- proofs of independence and conservativity;
- machine-checked theorems;
- a fixed semantics for the basic ontological predicates;
- validated domain interpretations.

The draft is therefore scientifically interesting and unusually precise for an emerging ontology, but it has not yet attained the same level of formal maturity as event structures, rewriting logic, Petri-net theory, or established modal logics.



## 16 Next Binding Research Programme

### 16.1 Phase A: Freeze the Signature

The aim is a complete list of types, functions, and relations, together with their arities, level membership, and intuitive semantics.

**Acceptance criterion:** No symbol in the core axioms remains untyped or explained only metaphorically.

### 16.2 Phase B: Non-Vacuous Minimal Model

A finite model is to be constructed with:

- at least two current configurations;
- at least one actual enactment;
- a typed direction that is open with respect to its target;
- a concrete trace field;
- a non-trivial continuity witness;
- qualitative novelty;
- sustained  $WF_{\text{ont}}$  as an explicitly interpreted model assumption.

**Acceptance criterion:** Every defining conjunction has an explicit witness; no core predicate is simply declared true across the board.

### 16.3 Phase C: Countermodels and Independence

For each core axiom, a model is to be sought that satisfies all the other axioms while violating the axiom in question.

**Acceptance criterion:** Redundant or unintentionally strong axioms are identified.

### 16.4 Phase D: Operational Representation

The core is represented in rewriting logic or in a coloured Petri net. Event structures supply the causality and conflict semantics; hypergraphs or structured cospans represent open multi-party enactments.

**Acceptance criterion:** The translation preserves types, actual event precedence, conflict, and the distinction between  $V_{\text{act}}$  and  $Adm_T$ .

### 16.5 Phase E: Isabelle/HOL Formalisation

The typed signature and minimal model are implemented in Isabelle/HOL. Initially, only syntactic well-formedness, satisfiability of the concrete model, and elementary preservation theorems are checked.

**Acceptance criterion:** Clean build without sorry or oops; documented Isabelle version, ROOT file, sources, and hashes.

### 16.6 Phase F: Module Tests

Every additional module receives:

1. a named formal gap;
2. a signature extension;
3. axioms or definitions;
4. a status as conservative, specialised, or interpretive;
5. a model or countermodel;
6. a clear statement of what the module does **not** prove.

## 16.7 Phase G: Domain Instances

Only after the core has been stabilised are biological, dissipative, quantum-physical, or social instances to be developed.

**Acceptance criterion:** Every instance has a typed interpretation of the core concepts and a preservation theorem for the respective claimed form of livingness.

## 17 Guidance for Scientific Review

Researchers should assess the draft by means of the following questions:

1. Is  $Offen_{\text{ont}}$  sufficiently grounded ontologically as a primitive predicate, or does it require an independent modal or causal semantics?
2. Are *Verankert* and *Realisierbar* sufficiently independent, or do their meanings overlap?
3. What minimal conditions must a continuity criterion  $c$  satisfy for  $K_c$  to be neither trivial nor overly restrictive?
4. Which trace types permit causal inference, and which permit only attributions of provenance or meaning?
5. Should  $Cfg$  be modelled as an attributed hypergraph, multiset, relational structure, or object of a monoidal category?
6. Which parts of the core can be formulated in first-order logic, and where is higher-order logic genuinely necessary?
7. Can  $Lebendig_{\text{fort}}$  be interpreted without a concealed totalisation of possible futures?
8. Which concrete countermodels distinguish transformation, creative transformation, and living enactment?
9. Is the asserted persistence of transformability a definition of livingness or an excessively strong selection criterion?
10. Which empirical or phenomenological cases distinguish the draft from competing process ontologies?

## 18 Conclusion

The living coherence is currently defined formally as **sustained, constrained formability at the configuration level**. Its core structure connects:

- a typed present configuration  $\Gamma$ ;
- a structured field of anchored directions  $\Pi_{\mathcal{Z}}(\Gamma)$ ;
- an actual enactment  $V_{\text{act}}$ ;
- an emerged configuration  $\Gamma'$ ;
- witnessed continuity  $K_c$ ;
- qualitative and, where applicable, ontological novelty;
- a typed trace field;
- sustained ontological transformability.

The theory has thereby advanced beyond the stage of a mere metaphor. Its conceptual architecture is sufficiently clear to be formalised, criticised, and tested by models. The decisive scientific reservation nevertheless remains: the theory represents ontological openness precisely, but has not yet proved its validity. It is precisely this distinction that makes the present state suitable for presentation and protects the project from the category errors identified above.

## Appendix A: Condensed Formal Core

$$\Pi_{\mathcal{Z}}(\Gamma) = \{d \mid P_{\mathcal{Z}}(\Gamma, d) \wedge Verankert_{\mathcal{Z}}(\Gamma, d) \wedge Realisierbar_{\mathcal{Z}}(\Gamma, d)\}.$$

$$WF_{\log}(\Gamma) \iff \exists d, \rho_1, \rho_2 [d \in \Pi_{\mathcal{Z}}(\Gamma) \wedge Adm_T(\Gamma, d, \rho_1) \wedge Adm_T(\Gamma, d, \rho_2) \wedge \rho_1 \not\equiv \rho_2].$$

$$WF_{\text{ont}}(\Gamma) \iff \exists d [d \in \Pi_{\mathcal{Z}}(\Gamma) \wedge Offen_{\text{ont}}(\Gamma, d)].$$

$$Wandel_c \iff V_{\text{act}} \wedge K_c \wedge Neu_{\text{qual}}.$$

$$SchoepferischerWandel \iff Wandel_c \wedge Neu_{\text{ont}}.$$

$$LebendigerVollzug \iff SchoepferischerWandel \wedge WF_{\text{ont}}(\Gamma').$$

$$Lebendig_{\min}(\mathcal{Z}) \iff \exists e, \Gamma, d, \Gamma' LebendigerVollzug(e, \Gamma, d, \Gamma').$$

$$\begin{aligned} Lebendig_{\text{fort}}(\mathcal{Z}) \iff & \exists \Gamma_0 Aktiv_{\mathcal{Z}}(\Gamma_0) \\ & \wedge \forall \Gamma [AktErreicht_{V_{\text{act}}}^*(\Gamma_0, \Gamma) \rightarrow WF_{\text{ont}}(\Gamma)] \\ & \wedge \exists e, \Gamma, d, \Gamma' LebendigerVollzug(e, \Gamma, d, \Gamma'). \end{aligned}$$

## Appendix B: Architectural Overview

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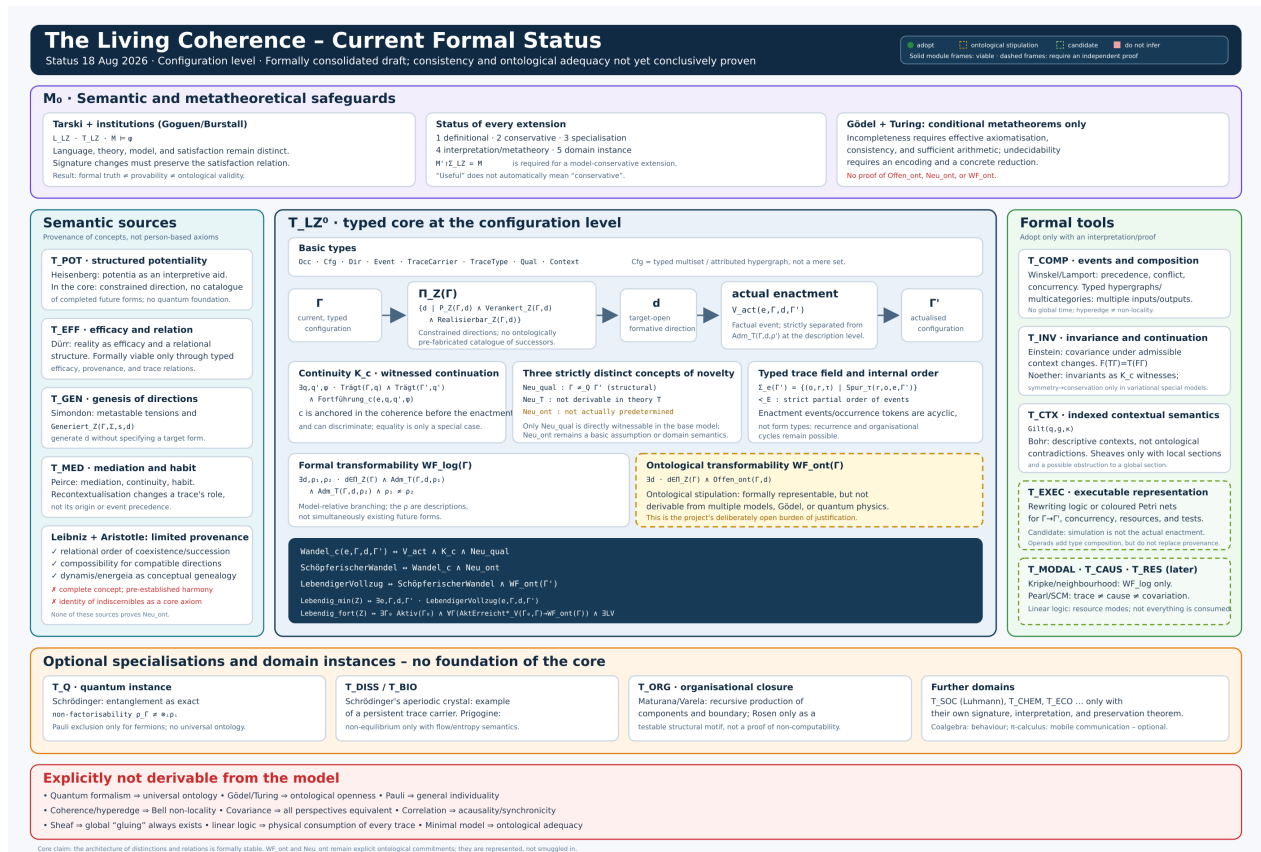


Figure 1: The Living Coherence – current formal status

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# The Living Coherence – Current Formal Status

Status 18 Aug 2026 · Configuration level · Formally consolidated draft; consistency and ontological adequacy not yet conclusively proven

● adopt    ▨ ontological stipulation    ▤ candidate    ■ do not infer  
Solid module frames: viable · dashed frames: require an independent proof

## M<sub>0</sub> · Semantic and metatheoretical safeguards

### Tarski + institutions (Goguen/Burstall)

$L_{LZ} \cdot T_{LZ} \cdot M \models \phi$   
Language, theory, model, and satisfaction remain distinct.  
Signature changes must preserve the satisfaction relation.  
Result: formal truth  $\neq$  provability  $\neq$  ontological validity.

### Status of every extension

1 definitional · 2 conservative · 3 specialisation  
4 interpretation/metatheory · 5 domain instance  
 $M' \models \Sigma_{LZ} = M$  is required for a model-conservative extension.  
“Useful” does not automatically mean “conservative”.

### Gödel + Turing: conditional metatheorems only

Incompleteness requires effective axiomatisation, consistency, and sufficient arithmetic; undecidability requires an encoding and a concrete reduction.  
**No proof of Offen\_ont, Neu\_ont, or WF\_ont.**

## Semantic sources

Provenance of concepts, not person-based axioms

### T\_POT · structured potentiality

Heisenberg: potentia as an interpretive aid.  
In the core: constrained direction, no catalogue of completed future forms; no quantum foundation.

### T\_EFF · efficacy and relation

Dürr: reality as efficacy and a relational structure. Formally viable only through typed efficacy, provenance, and trace relations.

### T\_GEN · genesis of directions

Simondon: metastable tensions and Generiert\_Z( $\Gamma, \Sigma, s, d$ )  
generate d without specifying a target form.

### T\_MED · mediation and habit

Peirce: mediation, continuity, habit.  
Recontextualisation changes a trace's role, not its origin or event precedence.

### Leibniz + Aristotle: limited provenance

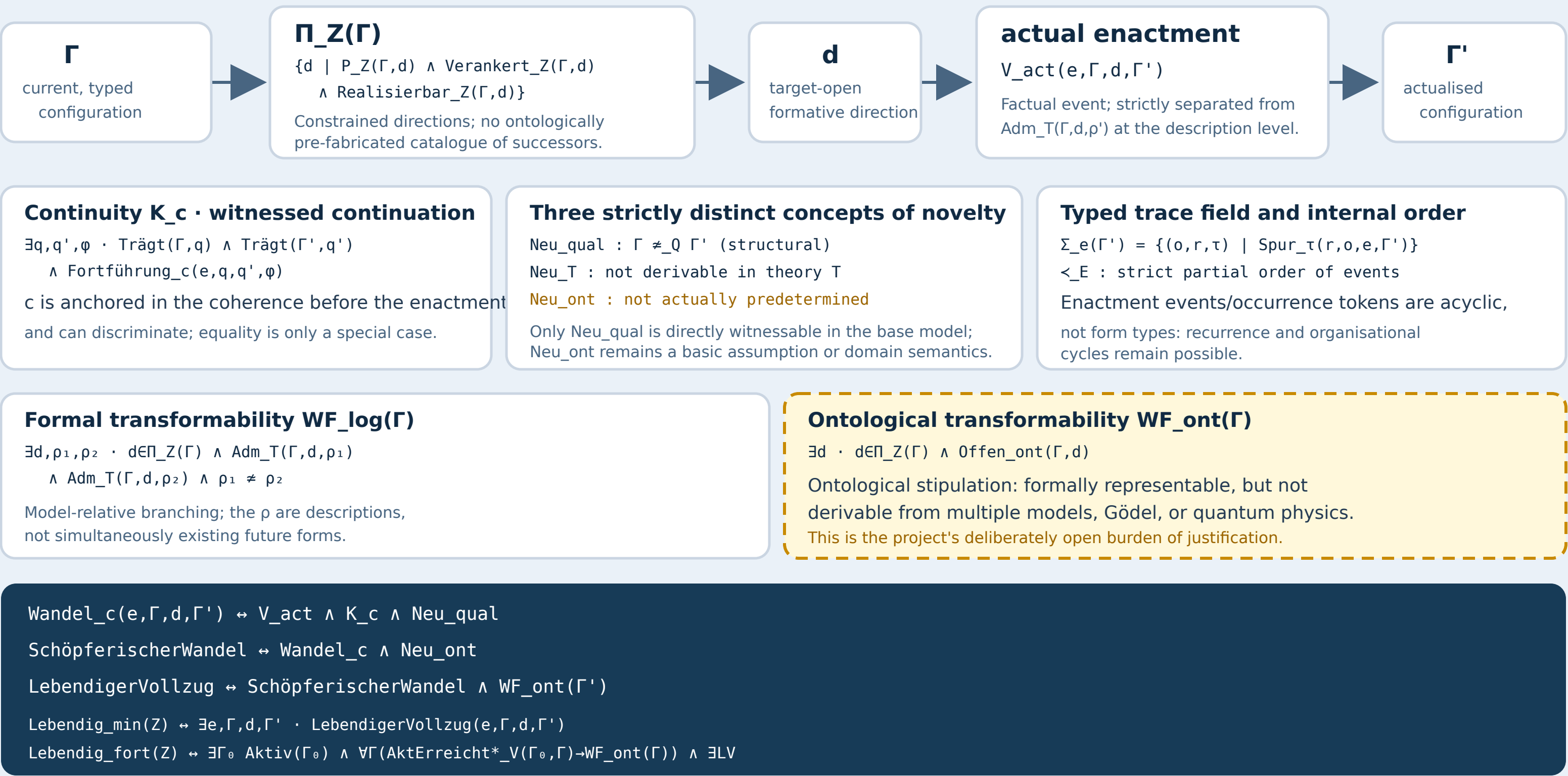
✓ relational order of coexistence/succession  
✓ compossibility for compatible directions  
✓ dynamis/energeia as conceptual genealogy  
✗ complete concept; pre-established harmony  
✗ identity of indiscernibles as a core axiom  
None of these sources proves Neu\_ont.

## T\_LZ<sup>0</sup> · typed core at the configuration level

### Basic types

Occ · Cfg · Dir · Event · TraceCarrier · TraceType · Qual · Context

Cfg = typed multiset / attributed hypergraph, not a mere set.



## Formal tools

Adopt only with an interpretation/proof

### T\_COMP · events and composition

Winskel/Lamport: precedence, conflict, concurrency. Typed hypergraphs/multicategories: multiple inputs/outputs.  
No global time; hyperedge  $\neq$  non-locality.

### T\_INV · invariance and continuation

Einstein: covariance under admissible context changes.  $F(T\Gamma) = T(F\Gamma)$   
Noether: invariants as K<sub>c</sub> witnesses;  
symmetry  $\rightarrow$  conservation only in variational special models.

### T\_CTX · indexed contextual semantics

Gilt( $q, g, \kappa$ )  
Bohr: descriptive contexts, not ontological contradictions. Sheaves only with local sections and a possible obstruction to a global section.

### T\_EXEC · executable representation

Rewriting logic or coloured Petri nets for  $\Gamma \rightarrow \Gamma'$ , concurrency, resources, and tests.  
Candidate: simulation is not the actual enactment.  
Operads add type composition, but do not replace provenance.

### T\_MODAL · T\_CAUS · T\_RES (later)

Kripke/neighbourhood: WF\_log only.  
Pearl/SCM: trace  $\neq$  cause  $\neq$  covariation.  
Linear logic: resource modes; not everything is consumed

## Optional specialisations and domain instances - no foundation of the core

### T\_Q · quantum instance

Schrödinger: entanglement as exact non-factorisability  $\rho_\Gamma \neq \otimes_i \rho_i$   
Pauli exclusion only for fermions; no universal ontology.

### T\_DISS / T\_BIO

Schrödinger's aperiodic crystal: example of a persistent trace carrier. Prigogine: non-equilibrium only with flow/entropy semantics.

### T\_ORG · organisational closure

Maturana/Varela: recursive production of components and boundary; Rosen only as a testable structural motif, not a proof of non-computability.

### Further domains

T\_SOC (Luhmann), T\_CHEM, T\_ECO ... only with their own signature, interpretation, and preservation theorem.  
Coalgebra: behaviour;  $\pi$ -calculus: mobile communication – optional.

## Explicitly not derivable from the model

- Quantum formalism  $\Rightarrow$  universal ontology · Gödel/Turing  $\Rightarrow$  ontological openness · Pauli  $\Rightarrow$  general individuality
- Coherence/hyperedge  $\Rightarrow$  Bell non-locality · Covariance  $\Rightarrow$  all perspectives equivalent · Correlation  $\Rightarrow$  acausality/synchronicity
- Sheaf  $\Rightarrow$  global “gluing” always exists · linear logic  $\Rightarrow$  physical consumption of every trace · Minimal model  $\Rightarrow$  ontological adequacy